

MANJIMUP, Australia

WMO: 946170

Lat: 34.2505S

Lon: 116.1450E

Elev: 287

StdP: 97.92

Time Zone: 8.00 (E08)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	5.0	1.8	4.4	11.7	2.7	4.8	10.3	11.0	14.2	9.8	14.1	2.8	20	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.8	33.3	19.0	31.1	18.5	28.9	18.0	20.6	28.7	19.7	27.4	18.9	26.2	4.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.6	22.1	17.4	12.9	21.2	16.6	12.3	20.6	60.5	28.5	57.4	27.5	54.8	26.1	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	7.0	DB	2.3	38.0	0.6	1.6	1.9	39.2	1.6	40.1	1.2	41.1	0.8	42.2	(4)
(5)			WB	2.0	22.5	0.7	1.3	1.5	23.5	1.1	24.3	0.7	25.0	0.2	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	19.9	20.2	18.8	16.0	13.6	11.5	10.5	10.9	11.7	13.7	16.4	18.2	(6)
(7)		DBStd	4.42	3.16	3.24	3.43	2.96	2.33	1.87	1.81	2.07	2.39	2.80	3.25	3.43	(7)
(8)		HDD10.0	48	0	0	0	0	2	6	16	14	9	2	0	0	(8)
(9)		HDD18.3	1446	18	16	35	82	148	205	243	231	200	146	77	44	(9)
(10)		CDD10.0	1903	306	286	272	182	113	51	31	41	59	118	191	254	(10)
(11)		CDD18.3	258	66	68	49	13	1	0	0	0	0	3	18	40	(11)
(12)		CDH23.3	2636	688	644	454	95	7	0	0	0	6	48	226	466	(12)
(13)		CDH26.7	1008	284	258	171	20	0	0	0	0	0	8	77	190	(13)
(14)	Wind	WSAvg	3.9	4.2	4.2	3.9	3.5	3.5	3.7	3.8	4.0	4.2	4.0	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	981	21	19	31	58	124	158	180	144	105	72	47	22	(15)
(16)		PrecMax	1269	92	60	125	194	259	278	289	242	184	143	122	77	(16)
(17)		PrecMin	697	0	1	4	4	48	14	102	50	38	9	6	3	(17)
(18)		PrecStd	136	24	16	26	39	51	60	54	44	37	34	30	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	35.1	35.1	29.4	24.2	19.2	18.1	20.2	23.7	27.7	32.6	35.5	(19)
(20)			MCWB	19.0	19.9	18.6	17.1	14.7	12.8	12.8	12.9	14.1	17.0	17.5	18.5	(20)
(21)		2%	DB	32.4	32.3	31.0	26.0	21.5	17.6	16.4	17.7	20.2	24.4	29.0	31.7	(21)
(22)			MCWB	19.1	19.2	18.6	16.9	14.2	12.9	12.7	12.9	13.8	15.7	16.9	18.4	(22)
(23)		5%	DB	29.8	29.9	28.0	23.5	19.4	16.2	15.1	16.1	17.9	21.5	25.8	28.3	(23)
(24)			MCWB	18.6	18.5	18.0	16.0	14.2	12.7	12.2	12.5	13.0	14.8	16.3	17.4	(24)
(25)		10%	DB	27.4	27.5	25.5	21.3	17.7	15.1	14.1	14.7	16.1	19.1	23.0	25.4	(25)
(26)			MCWB	17.9	18.0	17.0	15.5	13.8	12.3	11.6	11.8	12.4	13.8	15.4	16.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	22.2	20.7	19.1	17.4	15.4	15.0	15.4	16.0	18.2	19.2	20.9	(27)
(28)			MCDWB	29.9	30.2	28.5	24.2	19.9	16.8	16.0	17.1	19.9	24.7	28.2	30.0	(28)
(29)		2%	WB	20.5	20.6	19.7	18.0	16.0	14.3	13.7	14.0	14.7	16.6	18.0	19.2	(29)
(30)			MCDWB	28.5	28.5	27.1	22.2	18.8	15.9	15.2	15.9	18.4	22.4	25.4	28.3	(30)
(31)		5%	WB	19.5	19.5	18.8	17.2	15.2	13.5	12.7	13.1	13.7	15.4	17.0	18.1	(31)
(32)			MCDWB	27.1	27.0	25.0	21.4	17.7	15.1	14.3	15.3	16.9	20.1	23.8	26.3	(32)
(33)		10%	WB	18.5	18.6	17.9	16.3	14.4	12.7	11.9	12.2	12.7	14.3	16.0	17.0	(33)
(34)			MCDWB	25.8	25.6	23.3	20.1	16.9	14.4	13.6	14.4	15.6	18.1	22.2	24.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	12.8	11.6	9.7	8.4	7.2	7.2	7.7	8.7	10.0	11.4	12.3	(35)
(36)			MCDBR	18.0	17.4	16.2	13.5	10.9	8.8	8.6	9.6	11.8	14.5	16.8	18.0	(36)
(37)			MCWBR	6.4	5.9	5.6	5.5	5.5	5.5	5.3	5.4	6.0	6.7	6.6	6.5	(37)
(38)		5% WB	MCDWB	15.1	14.6	12.6	10.8	8.5	7.4	7.3	8.2	10.2	12.8	14.3	15.6	(38)
(39)			MCWBR	6.2	5.6	5.2	5.1	5.2	5.4	5.3	5.3	5.8	6.3	6.2	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.322	0.314	0.305	0.297	0.297	0.293	0.301	0.322	0.317	0.321	0.318	(40)	
(41)		taud	2.599	2.612	2.628	2.652	2.647	2.622	2.632	2.598	2.509	2.547	2.563	2.596	(41)	
(42)		Ebn at Noon	1014	995	967	923	878	849	873	913	942	988	1010	1023	(42)	
(43)		Edh at Noon	103	98	90	78	69	67	69	81	99	103	106	104	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.62	6.72	5.34	3.81	2.79	2.27	2.44	3.23	4.32	5.62	6.80	7.56	(44)	
(45)		RadStd	0.47	0.31	0.35	0.23	0.20	0.15	0.11	0.17	0.22	0.35	0.42	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-79	N/A	N/A

Nomenclature: See separate page